

Data-Driven Sales Analysis of Amazon Products Using Power BI

[PARVES BAPPI]

Introduction:

Power BI is Microsoft's business analytics platform that helps turn data into actionable insights. Nowadays, it is a valuable tool for analyzing data in the business area. This project aims to demonstrate practical experience using Power BI tools. While making this project, I used various Power BI toolkits to analyze my data, such as: ***Connecting to csv File, Data Cleaning, Data Modelling, Data Processing, Power Query, Date Tables, Time Intelligence Func, DAX, Date Function, Text Function, Filter Function, Calculate, YTD & QTD Functions, Custom Sorting, Formatting, Conditional Formatting, New Card Visual, Creating Charts, Formatting visuals, Creating Functions, Navigations and Dashboards***. Analyzing this data effectively can uncover valuable insights into customer preferences, product performance, and revenue trends.

Objective:

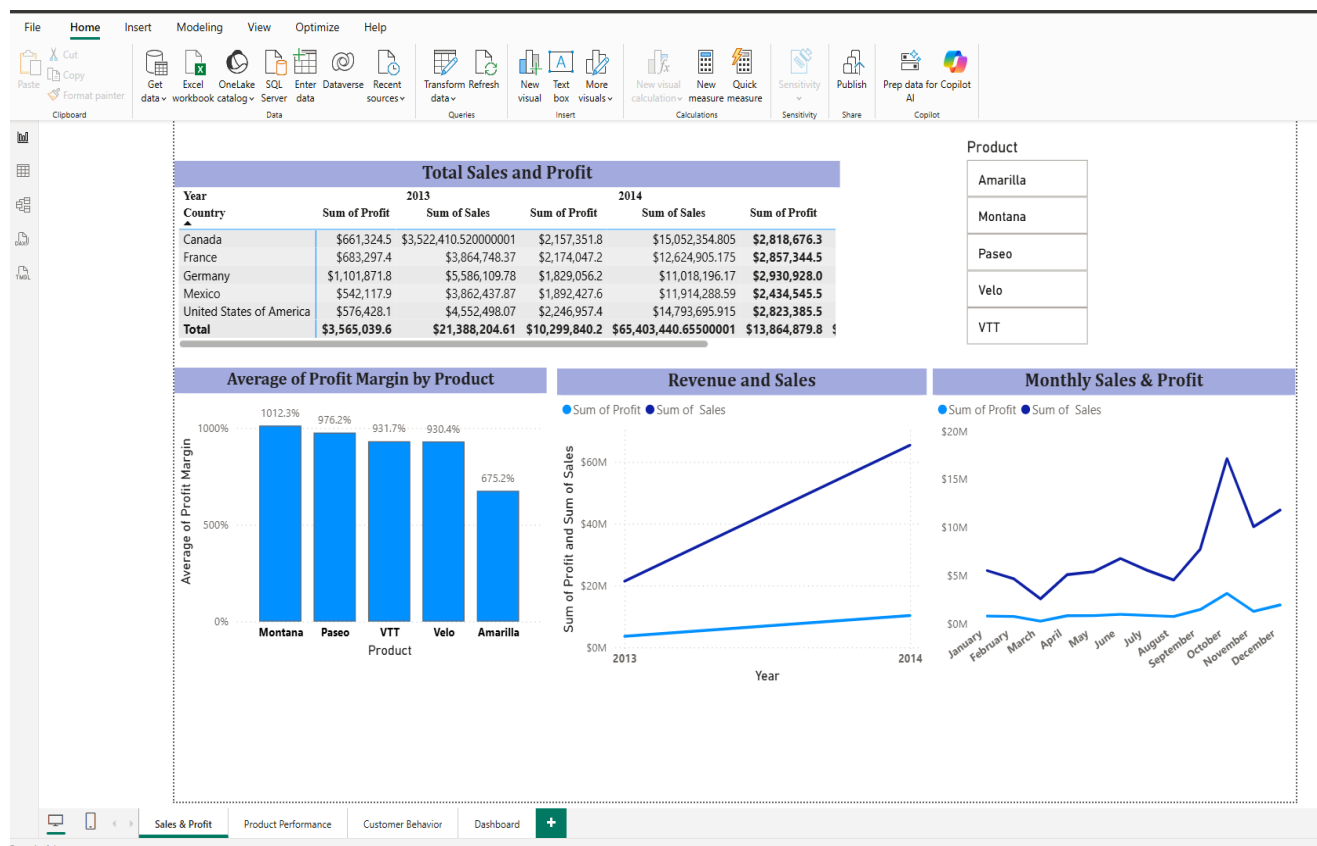
The objective of this project is to analyze Amazon product sales data using Power BI to identify key sales trends, top and least-performing products, and revenue drivers. By leveraging interactive dashboards, visualizations, and data-driven insights, the project aims to support informed decision-making, improve inventory management, and enhance overall sales performance. The insights derived from this analysis will not only enhance strategic planning but also help in optimizing inventory management, improving marketing strategies, and ultimately driving higher sales and profitability.

Data Collection:

I collected Amazon's sales (Q2) data from **Data.gov**, which is an open-source data platform. Additionally, I randomly chose customer names to enhance this project. After collecting these data, I connected to Excel files from Power BI and then performed some formatting, such as **Data Cleaning & ETL** (Extract, Transform, Load). More specifically, I filtered out profit values that were less than or equal to zero, unit prices less than \$2, and production costs less than \$2. Finally, this data is used to gain practical experience in Power BI, and any decisions or insights presented in this project do not have real-world connections.

Data Analysis:

First, I created a new column for Profit Margin under the Table View in the Financials data. The average profit margin ($\text{Profit/Sales} \times 100$) for Amazon is 618%, which shows a great value, although there are certain products that don't perform well, and Amarilla is one of them. Let's analyze the Profit and Sales performance of Amazon.



Page-1. Sales and Profit

There is a significant growth from 2013 to 2014 at Amazon that is shown at the first Matrix Table.

- ❖ **Total Sales:** Grew dramatically from \$21.39 million in 2013 to over \$65.40 million in 2014, indicating a nearly 200% increase.
- ❖ **Total Profit:** Also saw substantial growth, from \$3.57 million in 2013 to over \$13.86 million in 2014, a more than 280% increase. This suggests that profit grew at a faster rate than sales.

1.1. Country Performance:

- ❖ **United States** consistently leads in both sales and profit for both years.
- ❖ In 2013, the US had sales of \$4.55 million and profit of \$576,428.1.
- ❖ In 2014, US sales surged to \$14.79 million with profit reaching \$2.82 million.
- ❖ France and Germany are also major contributors, showing significant growth from 2013 to 2014.

The ranking of countries by sales and profit appears relatively consistent between the two years, with the US, France, Germany, and Canada being the top performers. Mexico consistently has the lowest sales and profit among the listed countries.

1.2. Average of Profit Margin by Product (Bar Chart)

The chart displays extremely high average profit margins, reinforcing the earlier statement about the project's data not reflecting reality (e.g., 1012.3% for Montana).

Top Performers:

- ❖ **Montana** has the highest average profit margin at 1012.3%.
- ❖ **Paseo** follows closely at 976.2%.
- ❖ **VTT** and **Velo** also show very high margins at 931.7% and 930.4% respectively.

Underperformer (as highlighted previously):

Amarilla has a significantly lower, yet still exceptionally high, average profit margin of 675.2% compared to other products. This confirms the earlier mention that Amarilla is one of the products that "can't perform well" in terms of profit margin relative to the others in this dataset.

1.3. Revenue and Sales (Line Chart)

This chart visually confirms the strong upward trend for both total profit and total sales from 2013 to 2014. Sales (blue line) are consistently higher than profit (orange line), as expected. The gap between sales and profit appears to widen in 2014, indicating that while profit is growing, sales volume is increasing even more rapidly, leading to a larger absolute difference between the two.

1.4. Monthly Sales & Profit (Line Chart - for 2014)

This chart provides a granular view of 2014's performance. Monthly sales and profit largely mirror each other, indicating a strong positive correlation between the two.

- ❖ **Seasonal Trends:** Both sales and profit show a clear upward trend towards the end of the year.
- ❖ **Peak Months:** Sales and profit seem to peak in October, November, and December, likely reflecting holiday season shopping. November appears to be the highest month for both.
- ❖ **Low Months:** The beginning of the year, particularly January and February, shows the lowest sales and profit figures. There's also a dip around July/August.

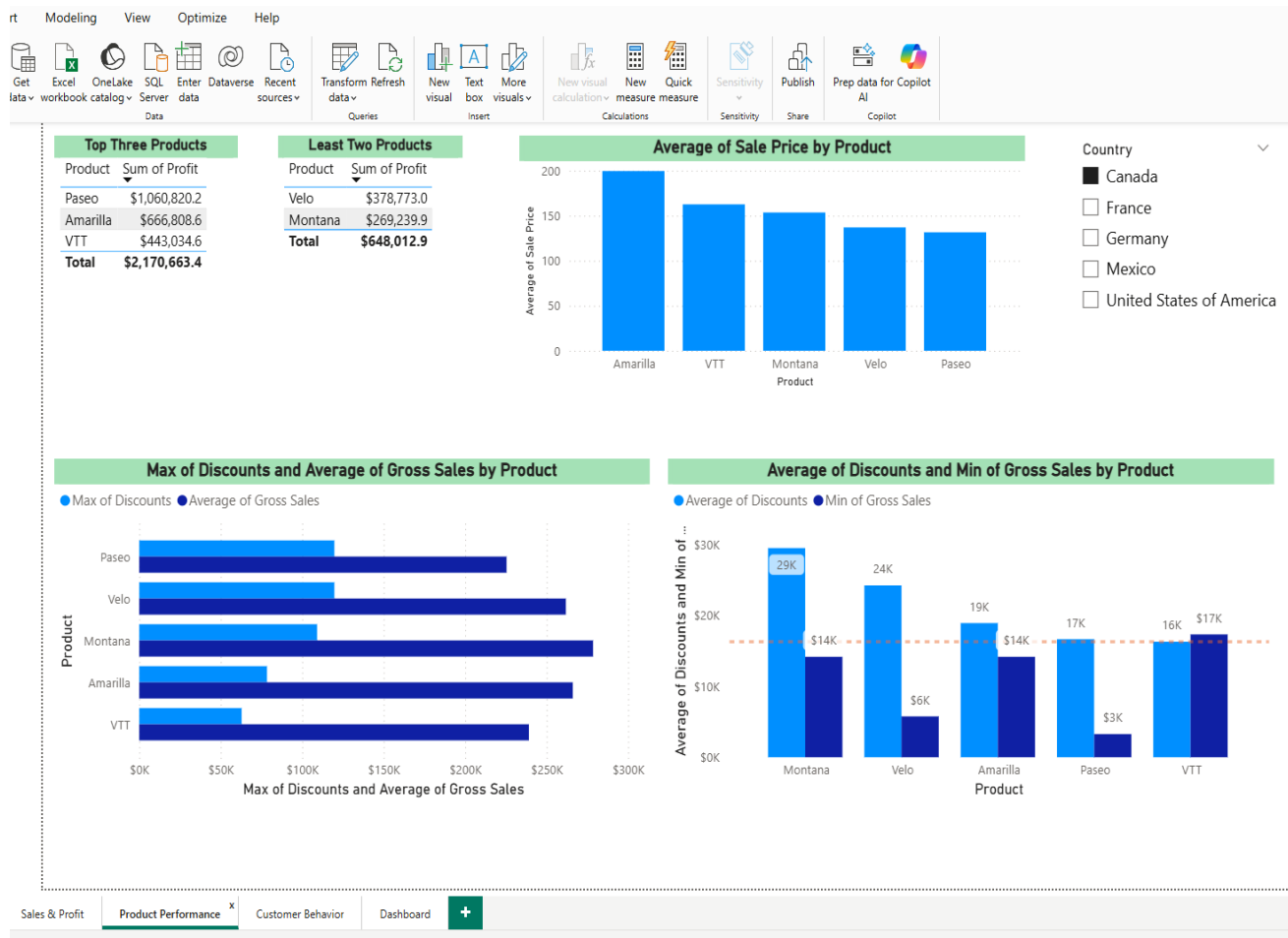
Page-2: Product Performance:

This page focuses on product-level performance, showcasing top and bottom products by profit, average sale prices, and the interplay of discounts and sales.

2.1. Top Three Products (Table)

- ❖ **Paseo** is the clear leader in terms of Sum of Profit, generating over \$1.06 million.
- ❖ **Amarilla** ranks second with a Sum of Profit of \$666,808.6.
- ❖ **VTT** is third with \$443,034.6 in profit.

The total profit for these top three products combined is substantial, at over \$2.17 million.



2.2. Least Two Products (Table)

- ❖ **Velo** has the lowest Sum of Profit at \$378,773.0.
- ❖ **Montana** is second to last with \$269,239.9 in profit.

The total profit for these two products is \$648,012.9, significantly less than the top single product, Paseo.

2.3. Average of Sale Price by Product (Bar Chart)

- ❖ **Amarilla** has the highest average sale price, close to \$200. This might explain its high overall profit despite a lower average profit margin percentage.
- ❖ **VTT** and **Montana Product** (likely Montana) follow, with average sale prices above \$150.
- ❖ **Velo** and **Paseo** have the lowest average sale prices among these products, both below \$150. This is interesting for Paseo, which is the top product by total profit, implying it achieves high profit through high volume rather than high individual sale price.

2.4. Max of Discounts and Average of Gross Sales by Product (Bar Chart - Top Left)

This chart compares the maximum discount given and the average gross sales for each product.

- ❖ **Amarilla** shows a very high average gross sales value (likely close to \$250K or more), and also a high maximum discount, suggesting it might be a product that sees large sales volumes and sometimes large individual discounts.
- ❖ **Montana** and **Velo** also have high average gross sales, with Montana appearing to have a higher maximum discount than Velo.
- ❖ **Paseo** and **VTT** have comparatively lower average gross sales values, which aligns with Paseo having a lower average sale price, but it still leads in total profit.

2.5. Average of Discounts and Min of Gross Sales by Product (Bar Chart - Top Right)

This chart highlights the average discounts applied and the *minimum* gross sales observed for each product.

- ❖ **Montana** and **Velo** show the highest average discounts (around \$29K and \$24K respectively).
- ❖ **Amarilla** has a significant average discount (around \$19K) but also a relatively high minimum gross sales value (around \$6K), indicating consistent sales even at their lowest points.
- ❖ **Paseo** and **VTT** have lower average discounts and lower minimum gross sales. Paseo's minimum gross sales appear very low (\$3K), reinforcing the idea of high volume driving its overall profit, but also potentially having periods of very low individual sales.

2.6. Country Slicer

The presence of a "Country" slicer (Canada, France, Germany, Mexico, United States of America) indicates that all the visuals on this dashboard can be filtered by country, allowing for country-specific product performance analysis.

Page-3: Customer Behavior

This dashboard provides insights into monthly sales performance, manufacturing cost distribution, and the relationships between units sold, sale price, discounts, and gross sales.

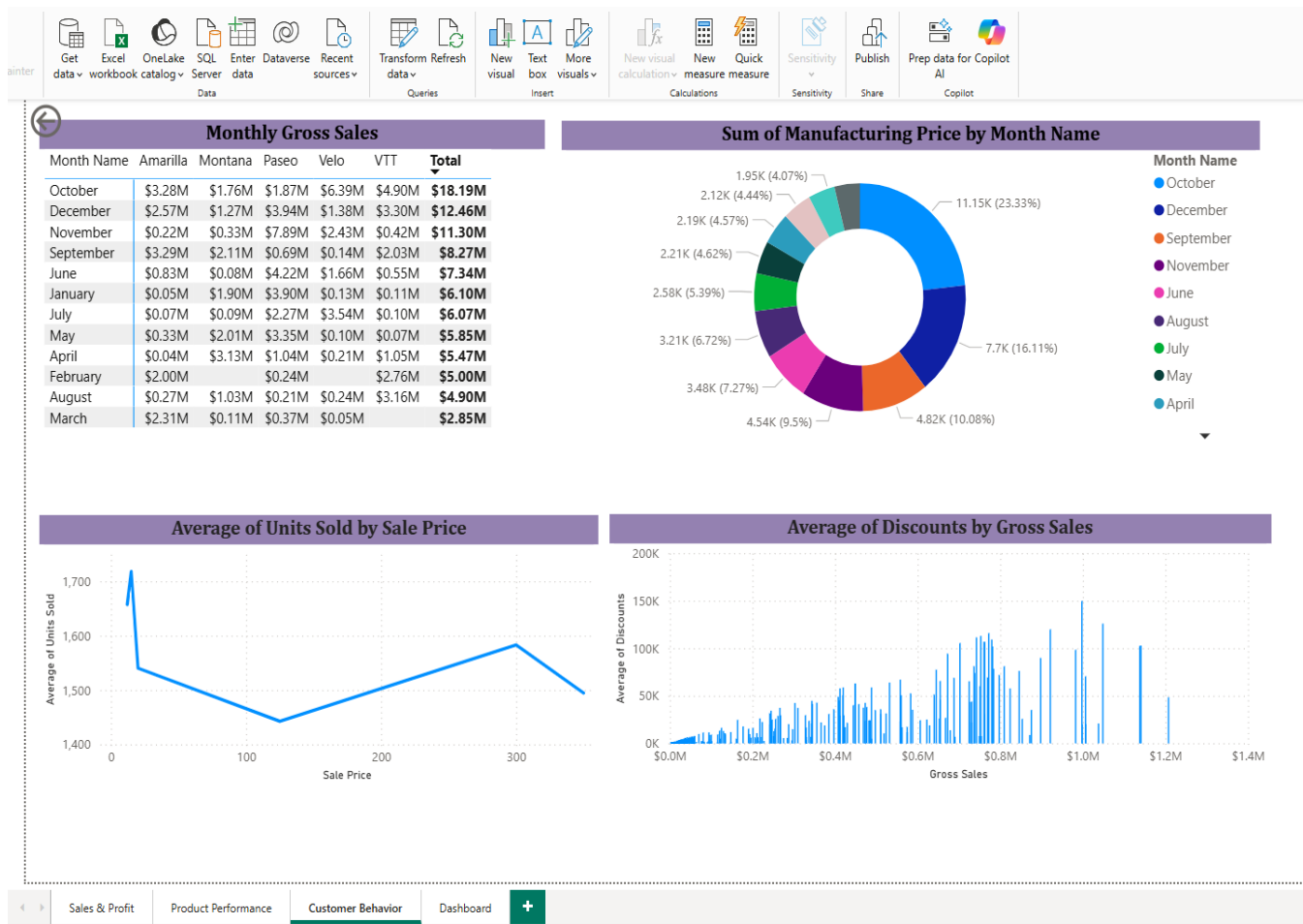
3.1. Monthly Gross Sales (Table)

- **Peak Sales Months: October** (\$12.46M), **December** (\$11.30M), and **November** (\$8.27M) are the months with the highest total gross sales. This aligns with the previous "Monthly Sales & Profit" chart which showed peaks towards the end of the year.
- **Lowest Sales Months: March** (\$2.85M), **February** (\$5.00M), and **August** (\$4.90M) show the lowest total gross sales.

3.2. Product Contribution

- **Paseo** consistently contributes a very high share of gross sales across most months, often being the top performer (e.g., \$6.39M in October, \$3.89M in December).

- **Amarilla** and **VTT** also show strong performance in peak months, with VTT especially strong in October (\$4.30M) and December (\$4.24M).
- **Montana** and **Velo** generally have lower individual monthly gross sales compared to the other three products.



3.3. Sum of Manufacturing Price by Month Name (Donut Chart)

This chart shows the distribution of the sum of manufacturing prices across different months.

- **October (23.33%)** and **December (16.11%)** account for the largest portions of total manufacturing price. This directly correlates with them being the months with the highest gross sales, suggesting higher production/sales volume in those months.
- **November (10.08%)** is also significant, aligning with its high sales.

The manufacturing price distribution generally mirrors the sales distribution, indicating that higher sales volume leads to higher associated manufacturing costs.

3.4. Average of Units Sold by Sale Price (Line Chart)

This chart depicts a general inverse relationship, common in demand curves, where a **higher sale price tends to correspond with a lower average of units sold**.

The highest average units sold (around 1,700) occur at lower sale prices (close to \$0). This is likely an artifact of the data cleaning (unit price less than \$2 was cut off), or very low-priced items. As the sale price increases, the average units sold generally decrease, though there are fluctuations. There's a significant drop in units sold after a certain price point, suggesting customer sensitivity to price. The curve flattens out at higher price points (e.g., beyond \$250), where units sold are consistently lower.

3.5. Average of Discounts by Gross Sales (Scatter Plot)

This chart explores the relationship between the average discount applied and the resulting gross sales. There appears to be a **positive correlation**: as gross sales increase, the average discount also tends to increase. This suggests that higher sales volumes might be achieved by offering larger discounts. Many data points cluster in the lower range of both discounts and gross sales, indicating many transactions/products with modest sales and discounts.

There are notable instances where very high average discounts (e.g., close to \$150K) are associated with some of the highest gross sales figures (e.g., \$1.0M - \$1.2M), reinforcing the idea of discounts driving large transactions. The spread of points suggests variability: some high gross sales are achieved with relatively low discounts, while others require substantial discounts.

Page-4: Dashboard

This is the main dashboard, acting as a high-level summary of Amazon's sales performance, designed for quick insights and interactive exploration. It consolidates key metrics and visualizations seen in previous tabs, making it the most critical overview. Let's break down its components for a deeper analysis:

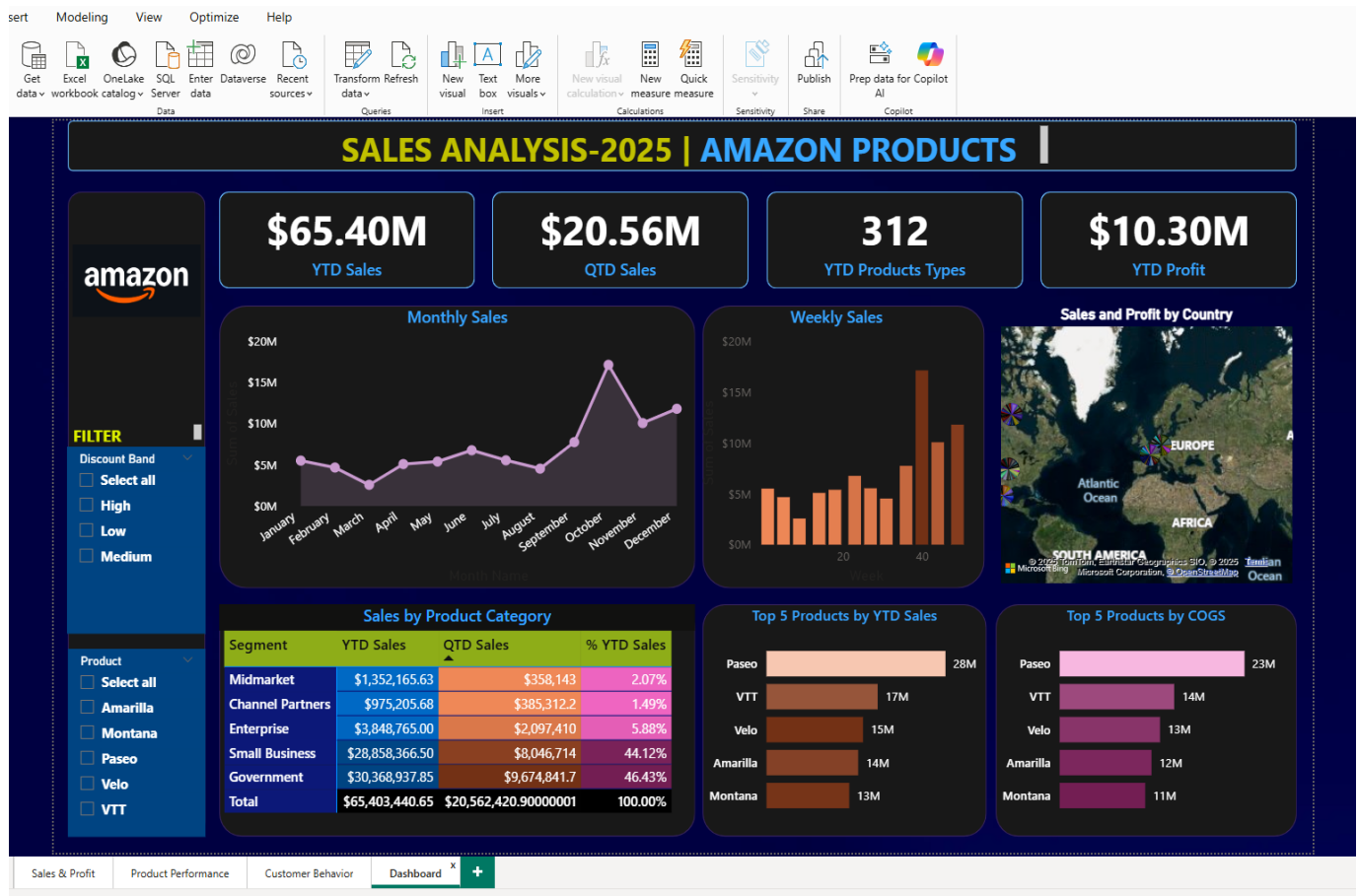
4.1. Key Performance Indicators (KPIs) at the Top

YTD Sales: \$65.40M: This represents the cumulative sales for the year to date. It aligns perfectly with the total sales observed for 2014 in the first dashboard, confirming that the "Year-To-Date" context likely refers to the full year 2014 in this project's dataset.

QTD Sales: \$20.56M: This shows the sales for the current quarter to date. Given the monthly sales trend peaking in Q4, this \$20.56M likely represents the sales for a significant portion of Q4 (October-December), or perhaps just December depending on the precise quarter definition. It indicates strong recent performance.

YTD Product Types-312: This metric shows the diversity of products sold year-to-date. A higher number suggests a wide product catalog being actively sold.

YTD Profit- \$10.30M: This is the total profit generated year-to-date. Comparing this to the YTD Sales (\$65.40M), the overall YTD profit margin for this dataset is approximately $\$10.30\text{M} / \$65.40\text{M} = 15.75\%$. While this is still a strong margin for a retail business, it's significantly lower than the extreme average profit margins (e.g., 618%, 1000%+) observed in the "Profit Margin by Product."



4.2. Monthly Sales (Line Chart):

This chart clearly visualizes the **strong seasonality** previously observed. Sales start low in January/February, show modest growth through spring/summer, and then dramatically **surge in** October, November, and December. November appears to be the peak month, reaching nearly \$20M in sales, followed closely by October and December. This reinforces the importance of the holiday shopping season for Amazon's sales within this dataset.

4.3. Weekly Sales (Bar Chart):

This chart provides an even more granular view of sales performance. It confirms the strong end-of-year surge, with the **last few weeks (likely corresponding to late Q4)** showing the highest sales volumes, with several weeks exceeding \$15M. The progression indicates steady growth throughout the year, culminating in explosive growth towards the end, consistent with the monthly trends.

4.4. Sales and Profit by Country (Map Visual):

The map visualizes sales/profit distribution geographically. Based on the intensity of color or size of the data points (though not perfectly clear without interaction), it appears:

- ❖ **North America (United States)** is the dominant region, confirming earlier table data.
- ❖ **Europe (France, Germany)** also shows significant contributions, consistent with the top-performing countries identified previously.

- ❖ Other regions, like South America and Africa, likely have much lower sales volumes within this dataset.

4.5. Sales by Product Category (Table):

This is a crucial insight into *what types* of products are driving sales. **Government** (\$30.36M YTD Sales, 46.43% of YTD Sales) and **Small Business** (\$28.85M YTD Sales, 44.12% of YTD Sales) segments collectively account for over 90% of all YTD sales. This is a profound insight, indicating that Amazon's sales are heavily concentrated in these two customer segments within this dataset. **Midmarket, Enterprise, and Channel Partners** contribute significantly less. Looking at QTD Sales, the trend holds, with Government (\$9.67M QTD) and Small Business (\$8.04M QTD) still dominating. This implies a consistent focus or strength in these B2B-like segments.

4.6. Top 5 Products by YTD Sales (Bar Chart):

Paseo is the top product by YTD sales (\$28M), confirming its consistent leading position across various metrics observed in the "Product Performance" dashboard (it was the top product by Sum of Profit). **VTT** (\$17M), **Velo** (\$15M), **Amarilla** (\$14M), and **Montana** (\$13M) follow. This visual quickly summarizes the best-selling individual products.

4.7. Top 5 Products by COGS (Cost of Goods Sold) (Bar Chart):

This chart indicates which products incur the highest costs for Amazon.

- ❖ **Paseo** also leads in COGS (\$23M). When compared to its \$28M YTD Sales, this implies a gross profit of \$5M for Paseo. This high COGS for Paseo is expected given its high sales volume.
- ❖ **VTT** (\$14M COGS vs. \$17M Sales) also shows a significant COGS.
- ❖ **Velo** (\$13M COGS vs. \$15M Sales), **Amarilla** (\$12M COGS vs. \$14M Sales), and **Montana** (\$11M COGS vs. \$13M Sales) follow.

The fact that the top 5 products by sales are almost identical to the top 5 products by COGS suggests that the products that sell the most naturally also incur the highest cost to procure or produce. The relative difference between sales and COGS for these products gives a quick visual indication of their gross profitability.

4.8. Filters/Slicers (Left Panel):

Discount Band (High, Low, Medium): This filter is crucial for deeper analysis. Users can select different discount bands to see how sales, profit, product types, and segment performance change. For instance, analyzing "High" discount band could reveal if heavily discounted products drive significant sales volume or if they erode profit margins too much.

Product (Specific Product Names): This allows users to drill down into the performance of individual products across the entire dashboard, providing a focused view on a single product's sales, profit, and contribution.

Recommendation:

- Given the overwhelming dominance of **Q4 sales** (October, November, December) as seen in both monthly and weekly sales charts, Amazon should double down on its resource allocation (marketing budget, inventory, logistics, staffing) for these months. **Implement "early-bird" holiday promotions starting in October to capture pre-peak demand.**
- **Prioritize and Promote "Paseo":** "Paseo" is consistently your top performer in both total profit and sales. Ensure it receives prominent placement on the platform, robust marketing support, and stable supply chain management. Consider bundling "Paseo" with complementary lower-performing products to boost their sales.
- **Intensify Focus on Government and Small Business:** Since these two segments account for over 90% of your YTD sales, dedicate more resources to understanding their unique needs. Enhance dedicated sales teams, offer tailored product catalogs, provide specialized customer support, and explore custom solutions or bulk purchasing incentives specific to Government and Small Business clients.
- Utilize the "Discount Band" filter more rigorously to understand the **profitability impact** (not just sales volume) of "High," "Medium," and "Low" discount tiers. Identify the optimal discount ranges that maximize *net profit* for different product categories and customer segments. Avoid across-the-board discounting; instead, apply discounts strategically to clear specific inventory, drive high-margin bundles, or acquire new customers.
- Continue to invest in and optimize operations (logistics, marketing, customer service) in your strongest markets, particularly the **United States, France, and Germany**. Explore opportunities for deeper market penetration or expanding your product offerings within these proven regions.
- While countries like Mexico show lower sales in this dataset, conduct external market research to evaluate their real-world growth potential. If justified, develop a phased market entry or expansion strategy, potentially starting with a limited product set or targeted marketing campaigns.

Conclusion:

This project successfully demonstrated the robust capabilities of Microsoft Power BI as a business analytics platform, effectively transforming raw data into actionable insights. The primary aim was to gain practical experience with various Power BI tools and methodologies, from data ingestion to sophisticated visualization and dashboard creation. Throughout this endeavor, a comprehensive range of Power BI toolkits were utilized. This project successfully served its aim of demonstrating practical experience with Power BI. It not only showcased the technical proficiency gained in data transformation, modeling, and visualization but also reinforced the critical thinking required to extract actionable insights from complex datasets, proving Power BI's invaluable role in modern business analytics.